

Data Flow Diagram and User Stories

Filled project document for GitHub and SkillWallet submission

Date	28 June 2026	Team ID	To be updated in SkillWallet
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction	Team Members	Devireddy Prasanna, Chokka Abitha, Navya Borugadda, Bodi Varshitha, Mulakala Harika Venkata Sridevi

Level 0 Data Flow

External Entity	Input Data	Process	Data Store	Output
Meteorologist / Officer	Rainfall, visibility, seasonal weather values	Flask route validates and preprocesses input	model.pkl and scaler.pkl	Flood or No Flood prediction
Data Scientist	Historical flood/weather dataset	EDA, preprocessing, train/test split, model training	Dataset, notebook, saved model	Accuracy, confusion matrix, selected model
Mentor / Analyst	Test records and project deliverables	Review model results and documentation	GitHub repository	Feedback and approval

User Stories

Epic	Story No.	User Story	Acceptance Criteria	Priority	Release
Data Collection	US-01	As a team member, I can collect historical flood/weather data.	Dataset has usable rainfall and flood label columns.	High	Sprint 1
Preprocessing	US-02	As a data analyst, I can clean and scale input features.	Missing values handled and train/test data prepared.	High	Sprint 1
Model Building	US-03	As a data scientist, I can compare Decision Tree, Random Forest, KNN, and XGBoost.	Best model selected using accuracy and classification report.	High	Sprint 2
Application	US-04	As an officer, I can enter weather parameters in a web form.	Valid input produces a flood risk result.	High	Sprint 3
Deployment	US-05	As a reviewer, I can access project files and demo from GitHub/SkillWallet.	GitHub and demo links submitted by Team Lead.	Medium	Sprint 4